GATE

ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 14



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Recap of Previous Lecture



Topic Boats & Streams, Trains

Topics to be Covered



Topic-1 Data Interpretation

[MCQ]

$T = \frac{D}{S}$ → 'x'

u

#Q. A boat traveled from A to B and back to A from B in 5 hours. If the speed of boat in still water and the speed of stream be 7.5 kmph and 1.5 kmph, then what is the distance between A and B?

Assignment

A 18 km

B 10 km

C 16 km

D 5 km

$\frac{x}{9} + \frac{x}{6} = 5$

$\Rightarrow \frac{5x}{18} = x \Rightarrow \frac{2x+3x}{18} = 5$

$\therefore x = 18$

Races

$\frac{A}{B} \times \frac{B}{C} = \frac{1000}{900} \times \frac{1000}{950}$ (50m)

$\Rightarrow \frac{A}{C} = \frac{1000}{855}$

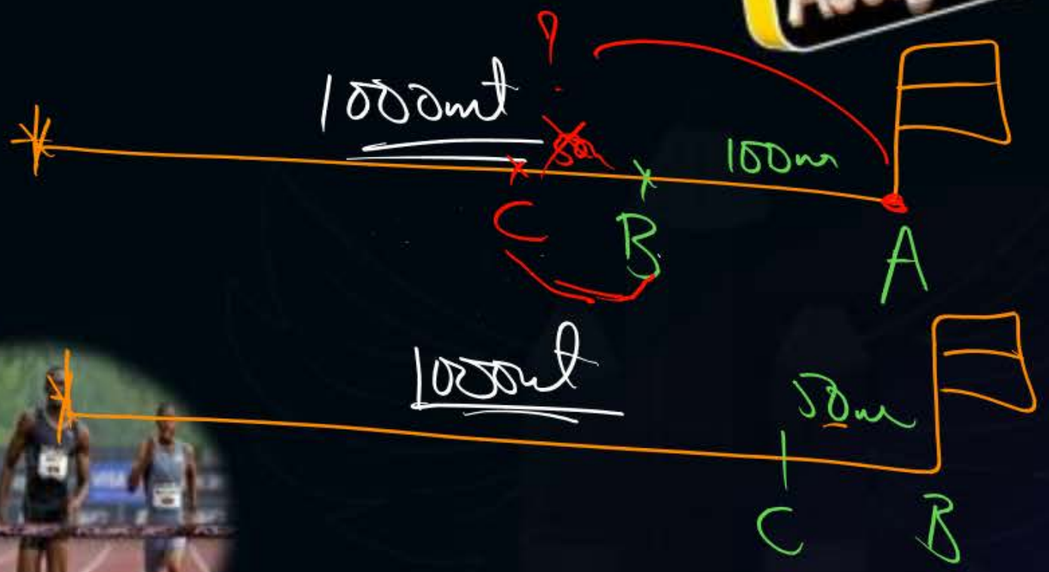
$1000 - 855 = 145m$

#Q. In a km race A defeats B by 100 meters and B defeats C by 50 meters, then A defeats C by how many meters?

$\frac{A}{B} = \frac{1000}{900}$

$\frac{B}{C} = \frac{1000}{950}$





Assignment

[MCQ]

#Q. In a km race A defeats (beats) B by 200 meters and B defeats C by 100 meters, then A defeats C by how many meters?

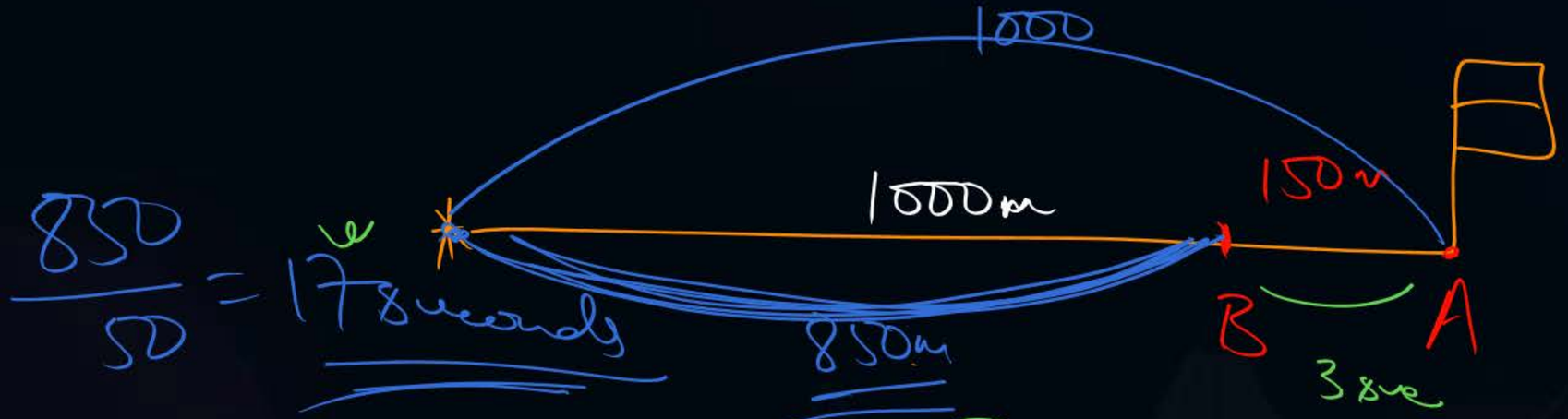
$$\frac{A}{C} = \frac{1000}{800} \times \frac{1000}{900}$$

$$\frac{A}{C} = \frac{1000}{720}$$

280 mts

[MCQ]

#Q. In a km race A defeats B by 150 meters or 3seconds. Find the speed of A. ?



$\frac{850}{50} = 17$ seconds

$v_A = \frac{1000}{17} = 58.82$ m/sec

$v_B = \frac{150}{3} = 50$ m/sec

[MCQ]

#Q. In a km race A defeats B by 200 meters or 5seconds whereas B defeats C by 100m. Find the speed of C.

$v_B = \frac{200}{5} = 40\ m/sec$

800

1000m

200m

B

$\frac{1000}{40} = 25\ seconds$

1000

5 sec

100

C

900

$v_C = \frac{900}{25} = 36\ m/sec$

Data Interpretation

Graph

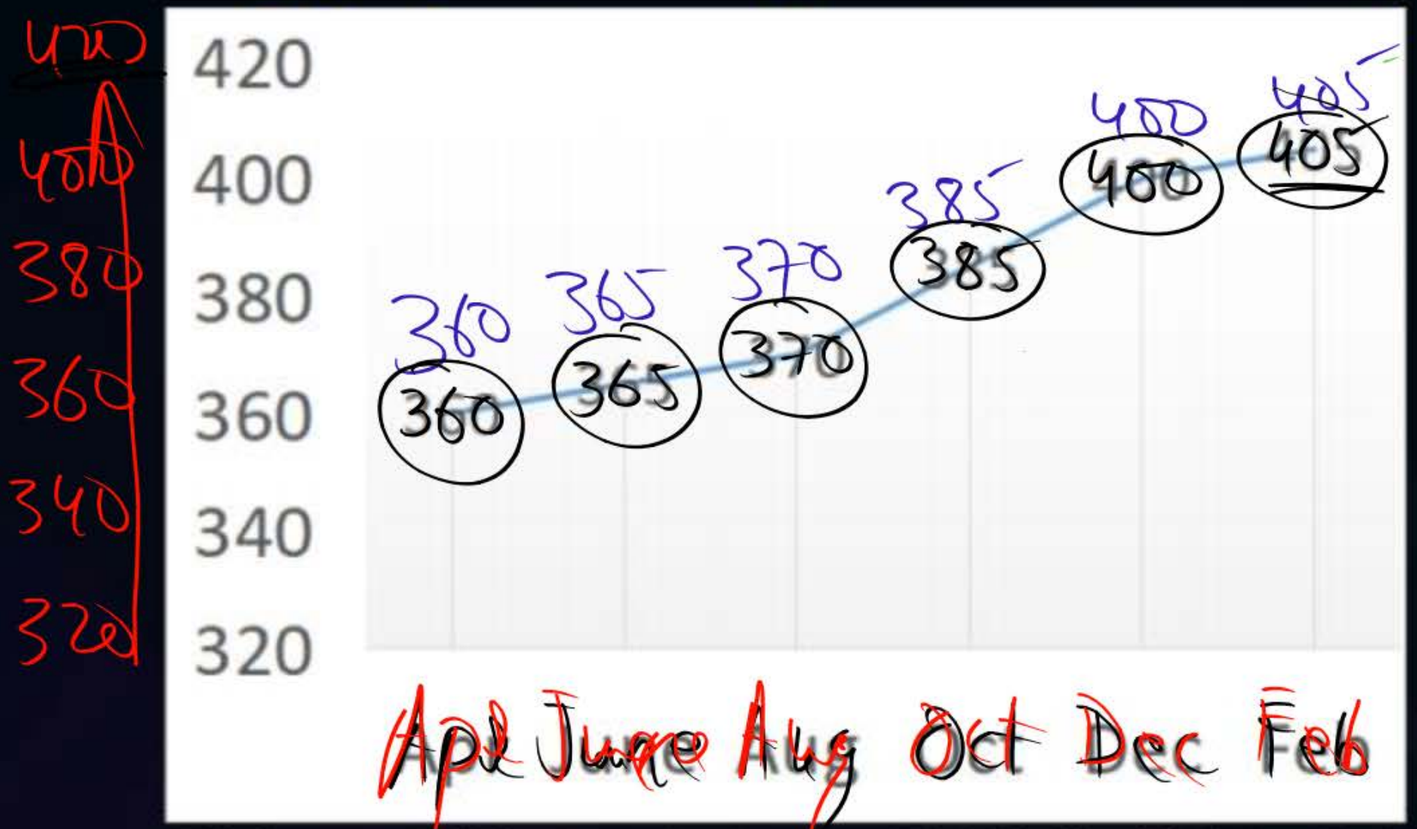
Types:

The numerical data pertaining to any event can be presented by any one or more of the following methods.

1) Tables
2) Line Graphs
3) Bar Graphs or Bar Charts
4) Pie Charts or Circle Graphs

Line Graph:

Study the following graph carefully and answer the questions:



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]

#Q. What is the percentage of marks scored by the student in the unit test of August and October taken together?

1000 { Aug ⇒ 370
Oct ⇒ 385
755

A 50%

B 83.3%

C 60%

D 75.5%

$\frac{755}{1000} \times 100$

$= \frac{755}{10}$

$= 75.5\%$



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

370 385

[MCQ]

$\frac{2285}{6} = 380.8\overline{3}$

#Q. What are the average of marks obtained by the student in the unit test of given academic year?

A. 361.6

B. 380.8

C. 377.5

D. 400.0

725 Apr ⇒ 360
June ⇒ 365

735 Aug ⇒ 370
Oct ⇒ 385

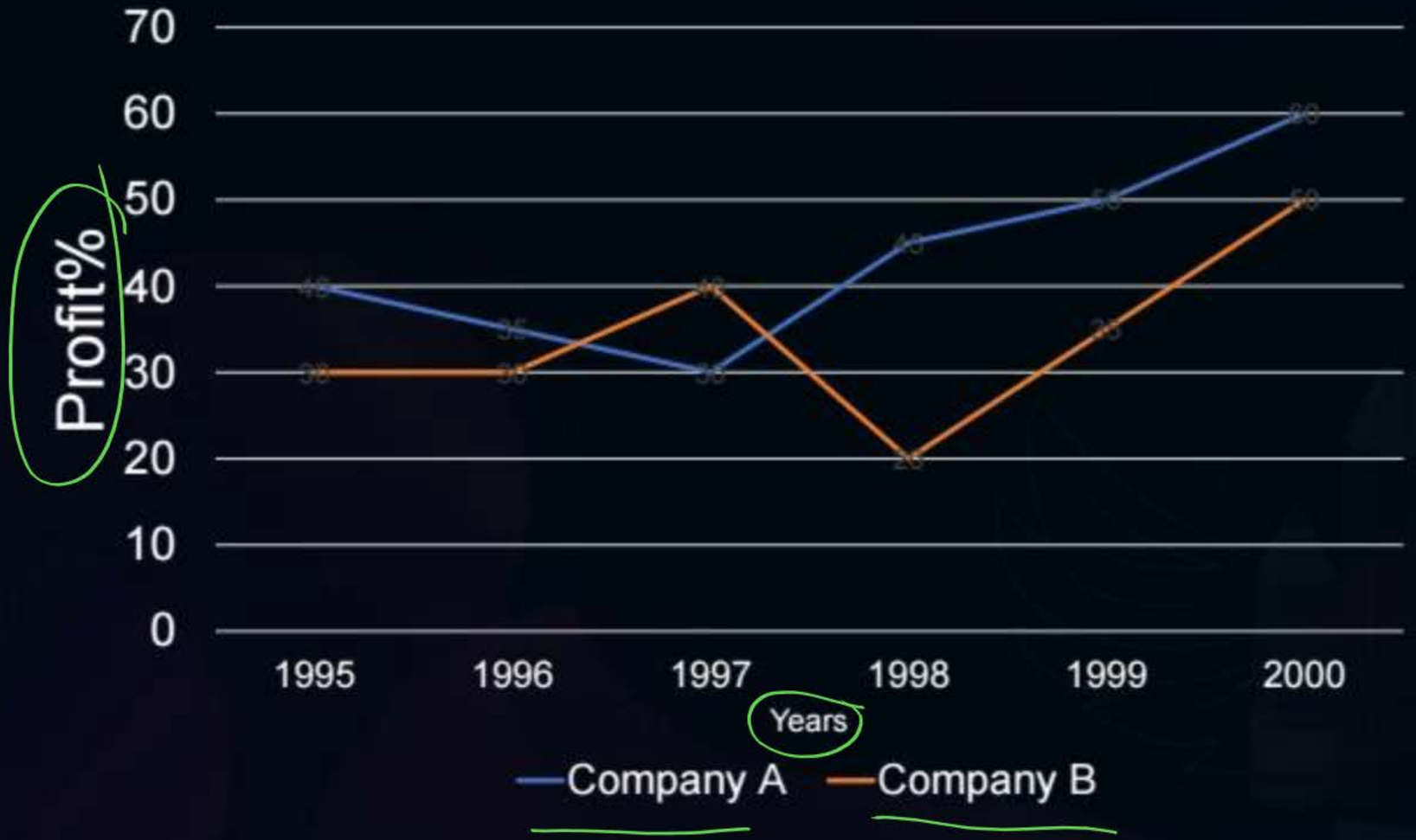
805 Dec ⇒ 400
Feb ⇒ 405

2285



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

Study the following table and answer the questions:



[MCQ]

#Q. If the expenditure of the company B in the year 1998 was ₹200 crore, what was its income?

P = 20%

$\frac{S.P}{C.P} = 1.2$

$\frac{Income}{Exp} = 1.2$

$\frac{Income}{200} = 1.2$

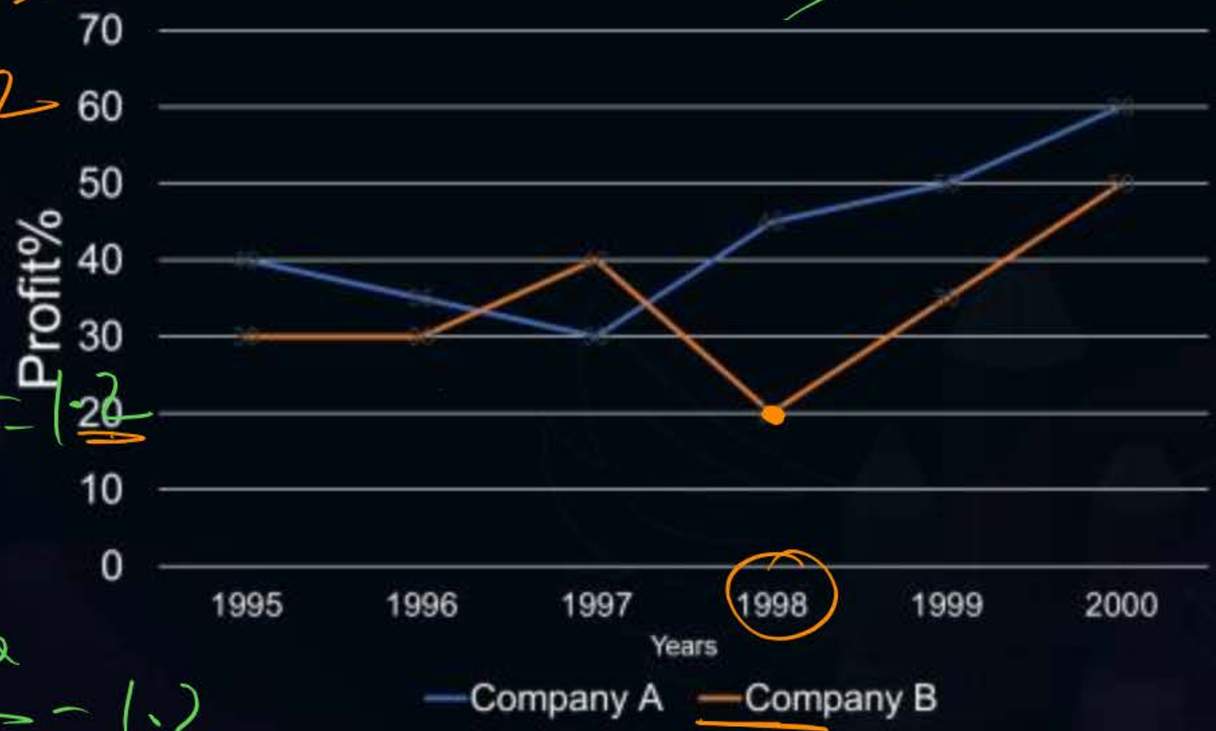
⇒ Income = 200 × 1.2 = 240 crores

A 360 crores

B 300 crores

C 240 crores

D 120 crores



[MCQ]

#Q. If the income of company A in the year 2000 was ₹600 crore, what was its expenditure?

P = 60%

∴ Expenditure = 375 Crore

$\frac{\text{Income}}{\text{Exp}} = 1.6$

A 210 crores

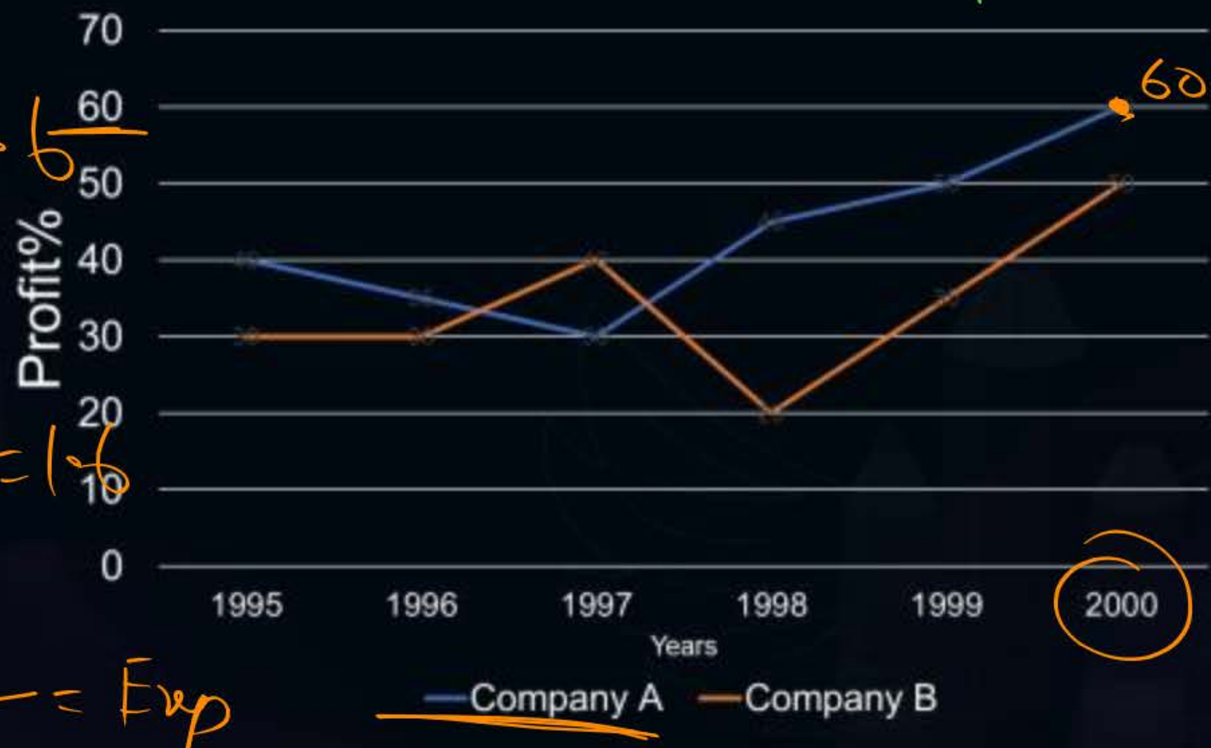
B 275 crores

$\Rightarrow \frac{600}{\text{Exp}} = 1.6$

C 340 crores

D 375 crores

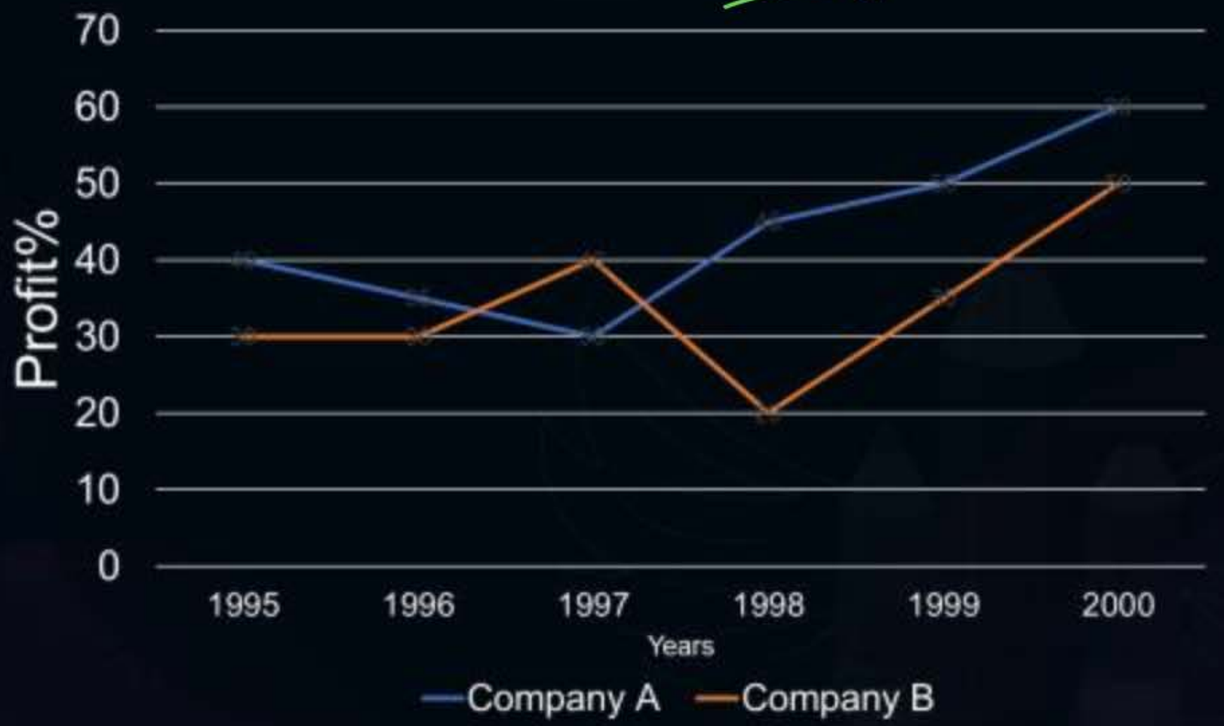
$\Rightarrow \frac{600}{1.6} = \text{Exp}$



[MCQ]

#Q. If the income of the two companies were equal in 1996, what was the ratio of their expenditure?

Assignment



A 26 : 27

B 100 : 67

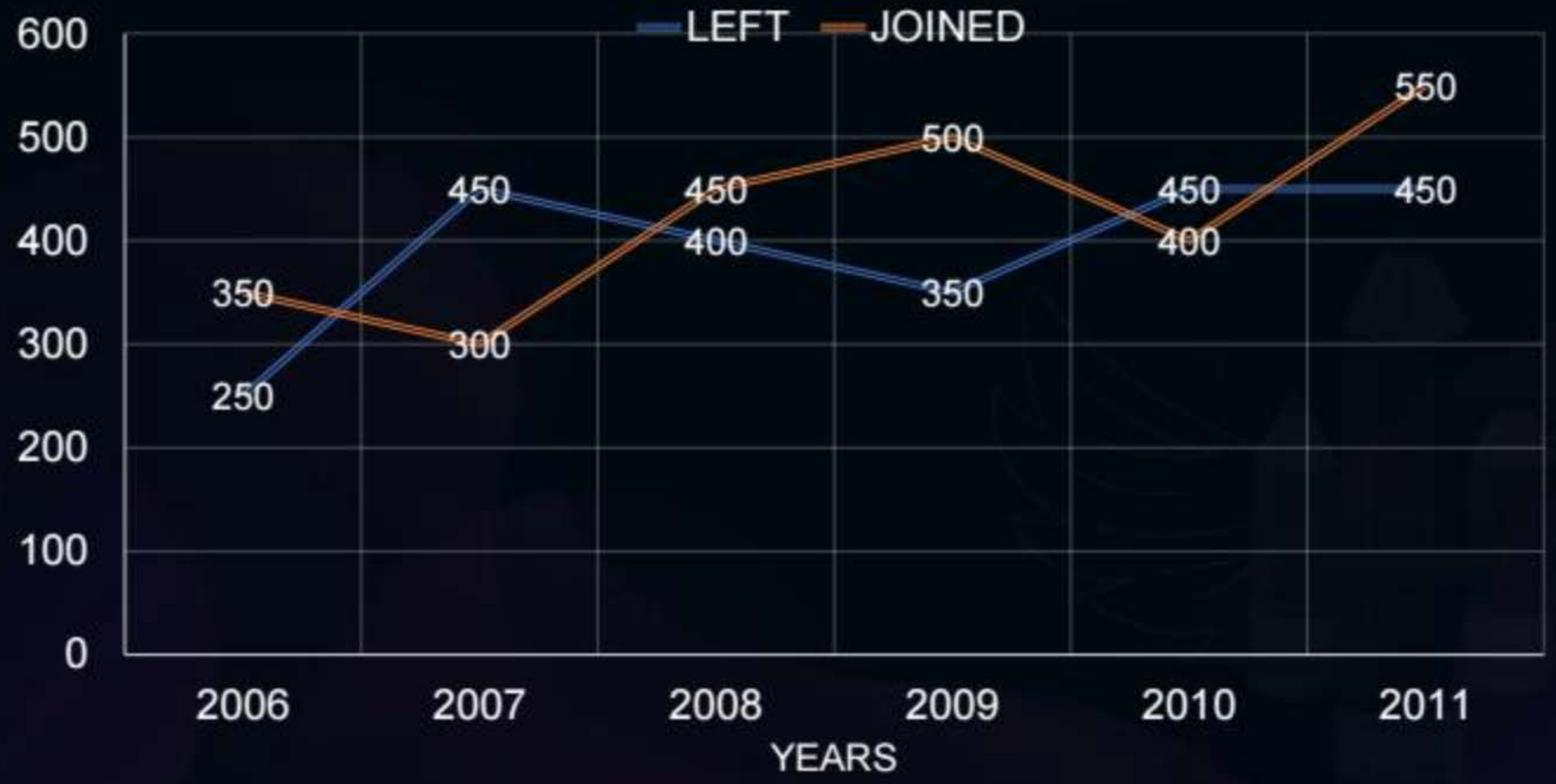
C 1 : 2

D 5 : 27

The following graph depicts the number of students who joined and left the school in the beginning of the academic year.

Assignment

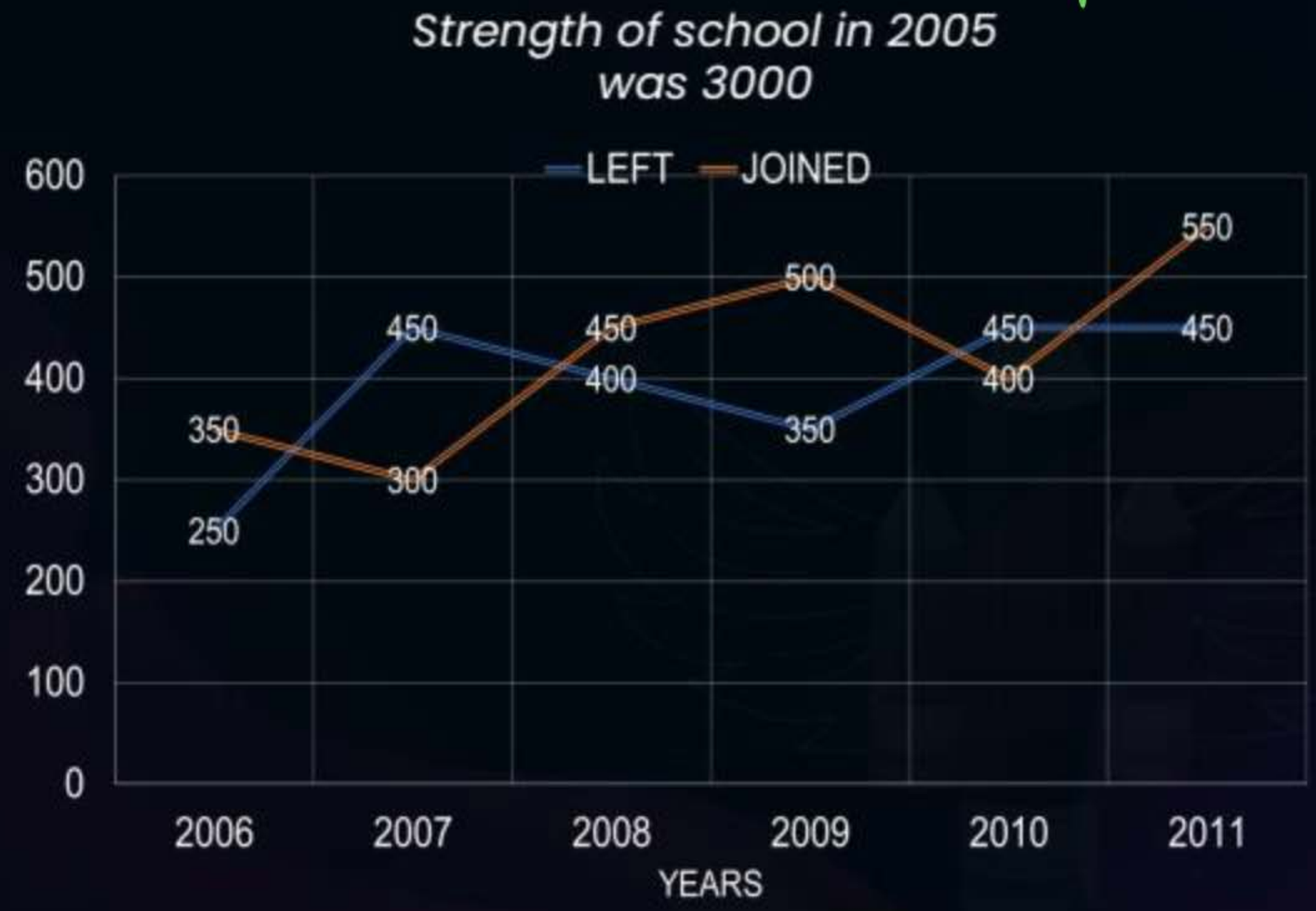
Strength of school in 2005 was 3000
2005 में स्कूल की संख्या 3000 थी



[MCQ]

#Q. By approximately what percent, the strength of school increased or decreased from 2007 to 2008?

Assignment



A 1.2%

B 1.7%

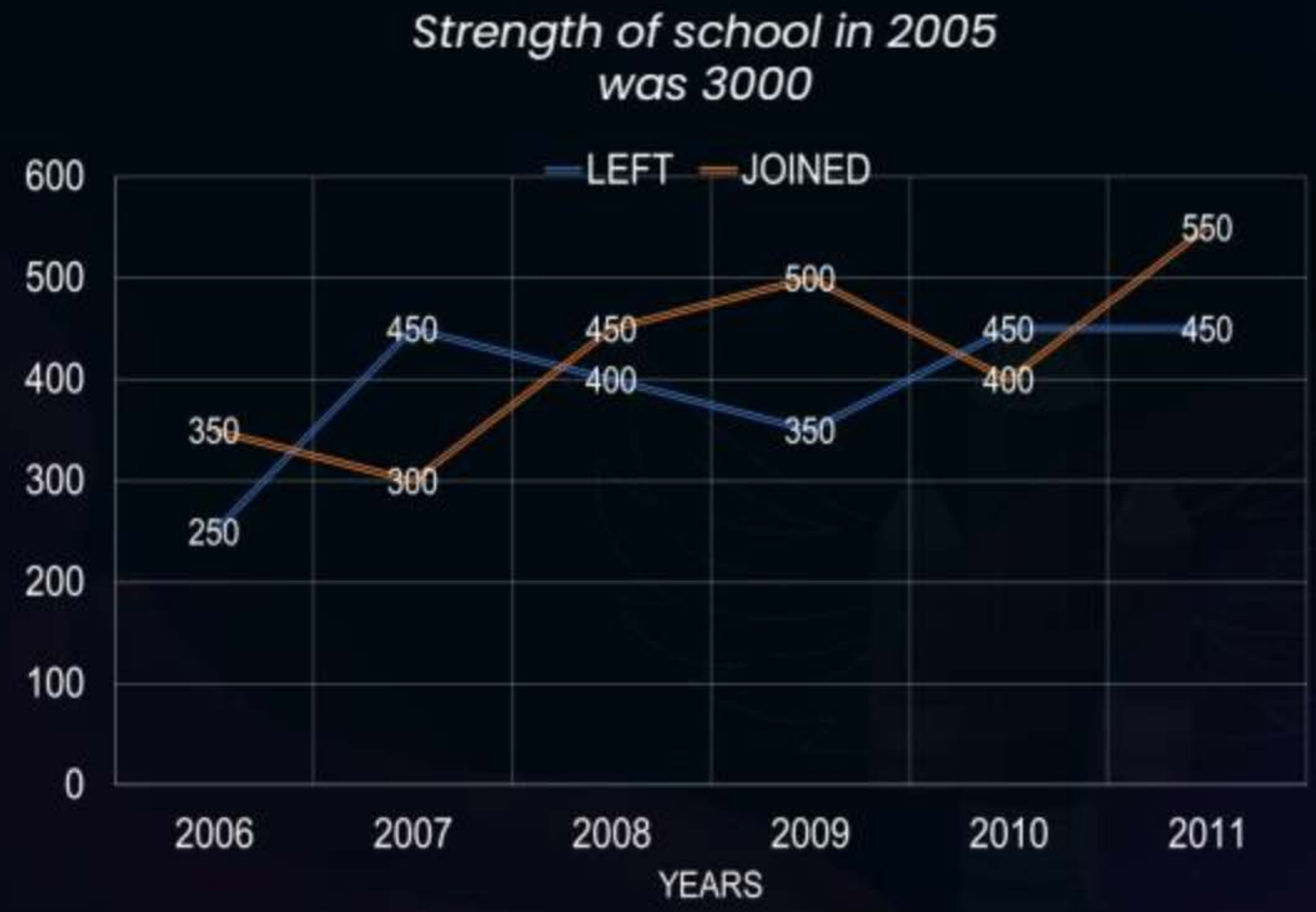
C 2.1%

D 2.4%

[MCQ]

#Q. How many students were studying in the school during 2009?

Assignment



A 2950

B 3000

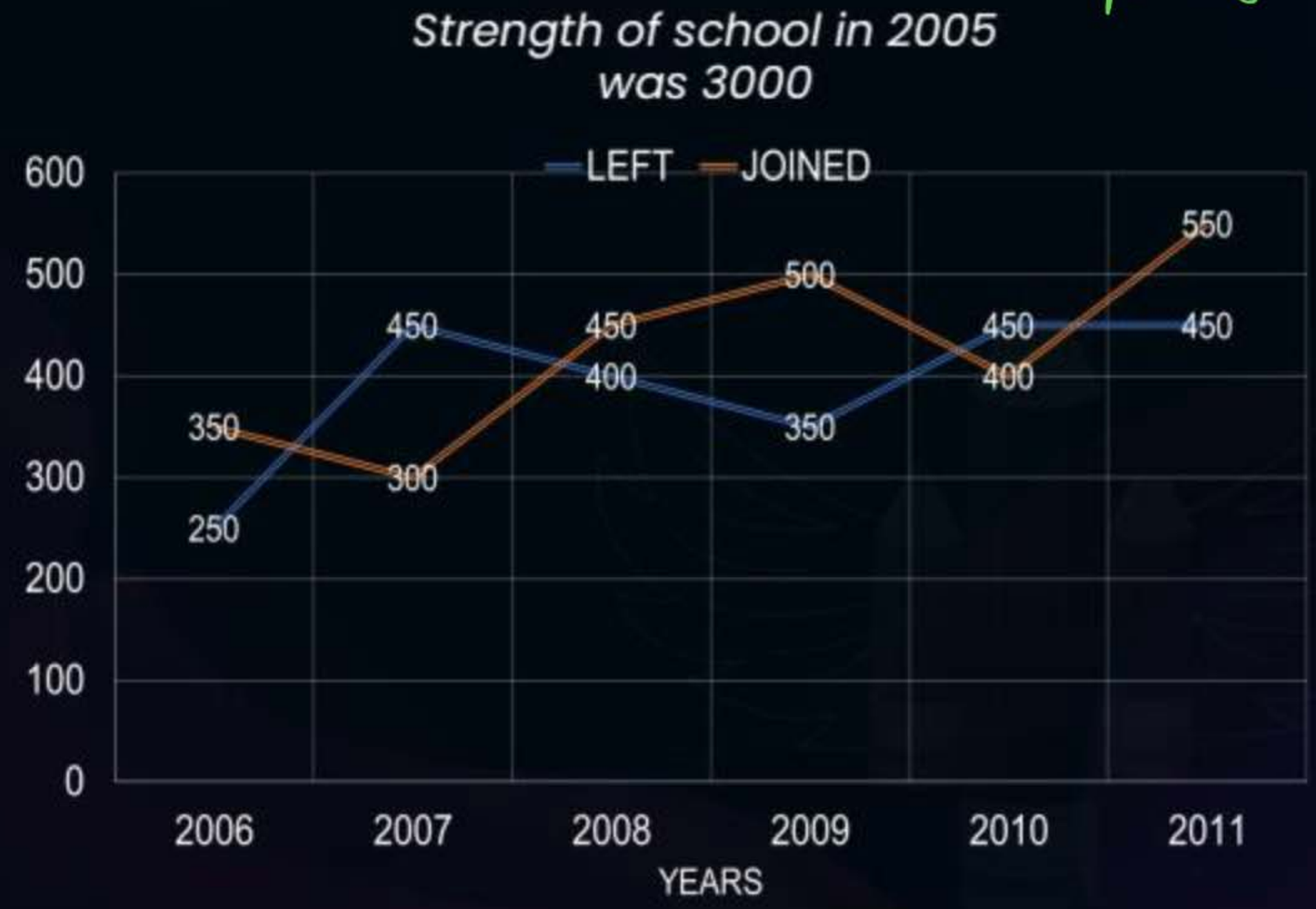
C 3100

D 3150

[MCQ]

#Q. The number of students studying in school in 2008 was what percent of the number of students studying in the scool in 2011?

Assignment



A 92.13%

B 93.75%

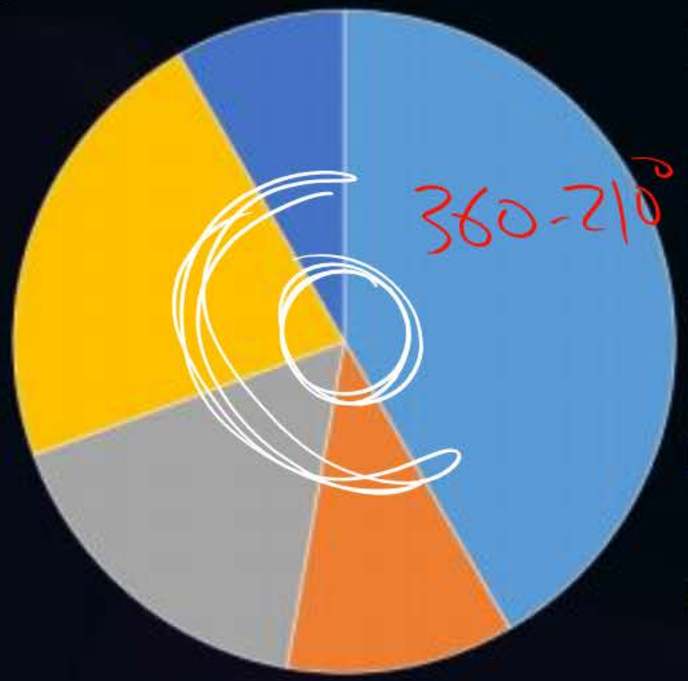
C 97.25%

D 96.88%

Pie Chart:

Study the following graph carefully and answer the questions:

POPULATION OF 24000



- UNEMPLOYED → ? → 150°
- SELF EMPLOYED - 40°
- PUBLIC SECTOR - 60°
- PRIVATE SECTOR - 80°
- GOVT.SECTOR - 30°

360 - 210°

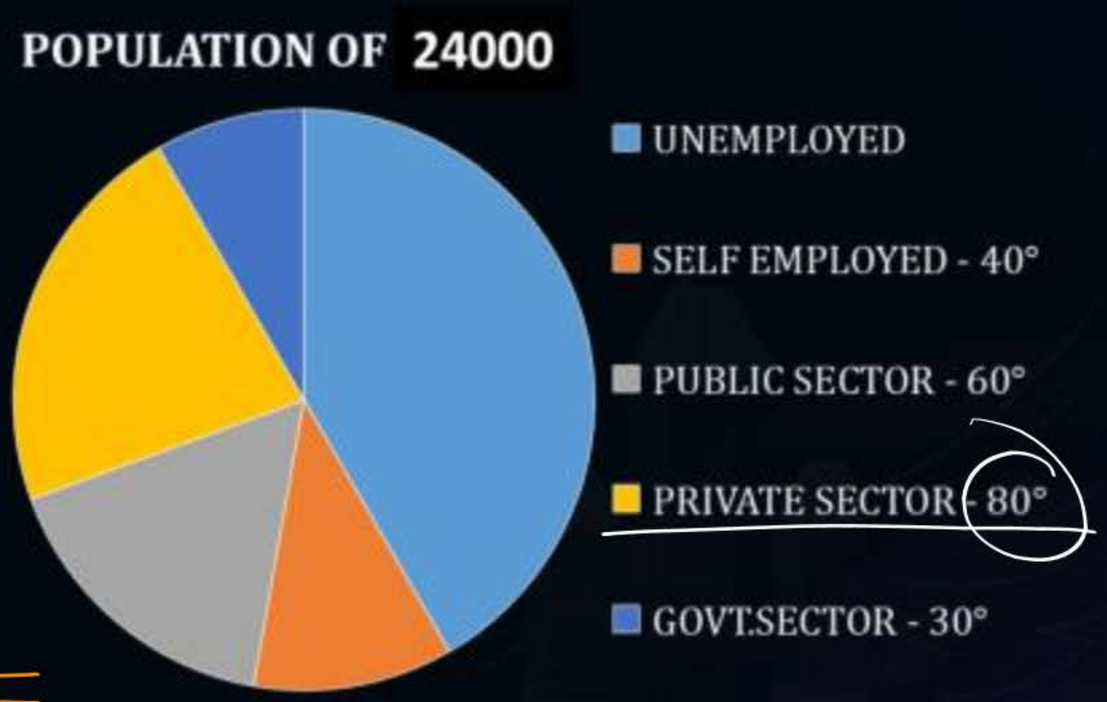
210°

[MCQ]

#Q. If private sector employees are to be mentioned in ~~above~~ below chart in the form of percentage, it would be .....%.

$$\frac{80^\circ}{360^\circ} \times 100$$

$$= \frac{250}{9}\% = 22.\overline{2}\%$$



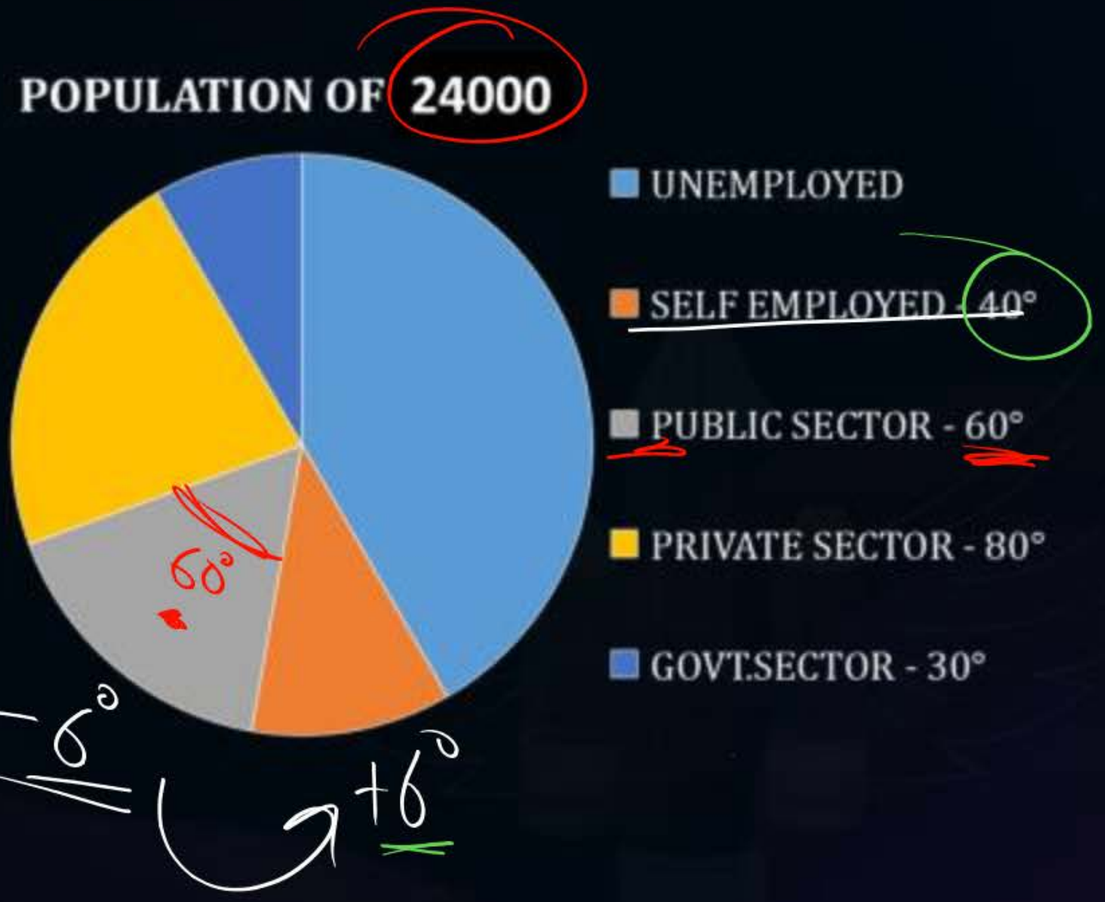
[MCQ]

#Q. If 10% of public sector people resign and becomes self-employed, find the increase percent in self-employed population.

$\frac{6^3}{40} \times 100$

$\frac{40}{2}$

$= \frac{30}{2} = 15\%$



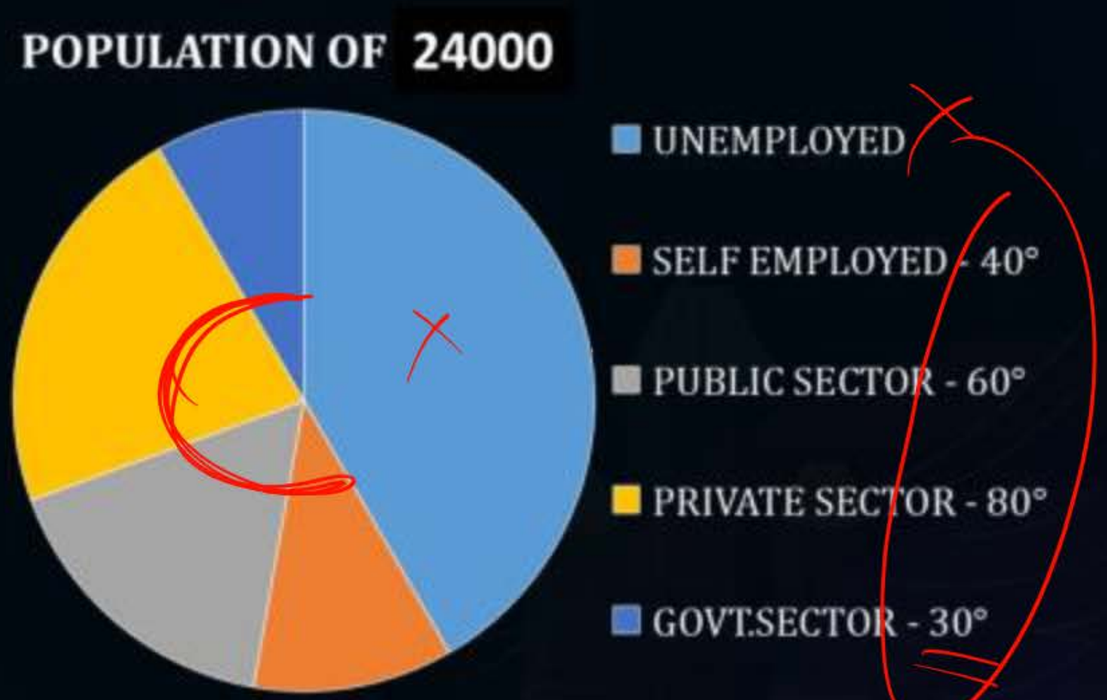
[MCQ]

#Q. What percentage of employed people are working in government sector?

$$\frac{30}{210} \times 100$$

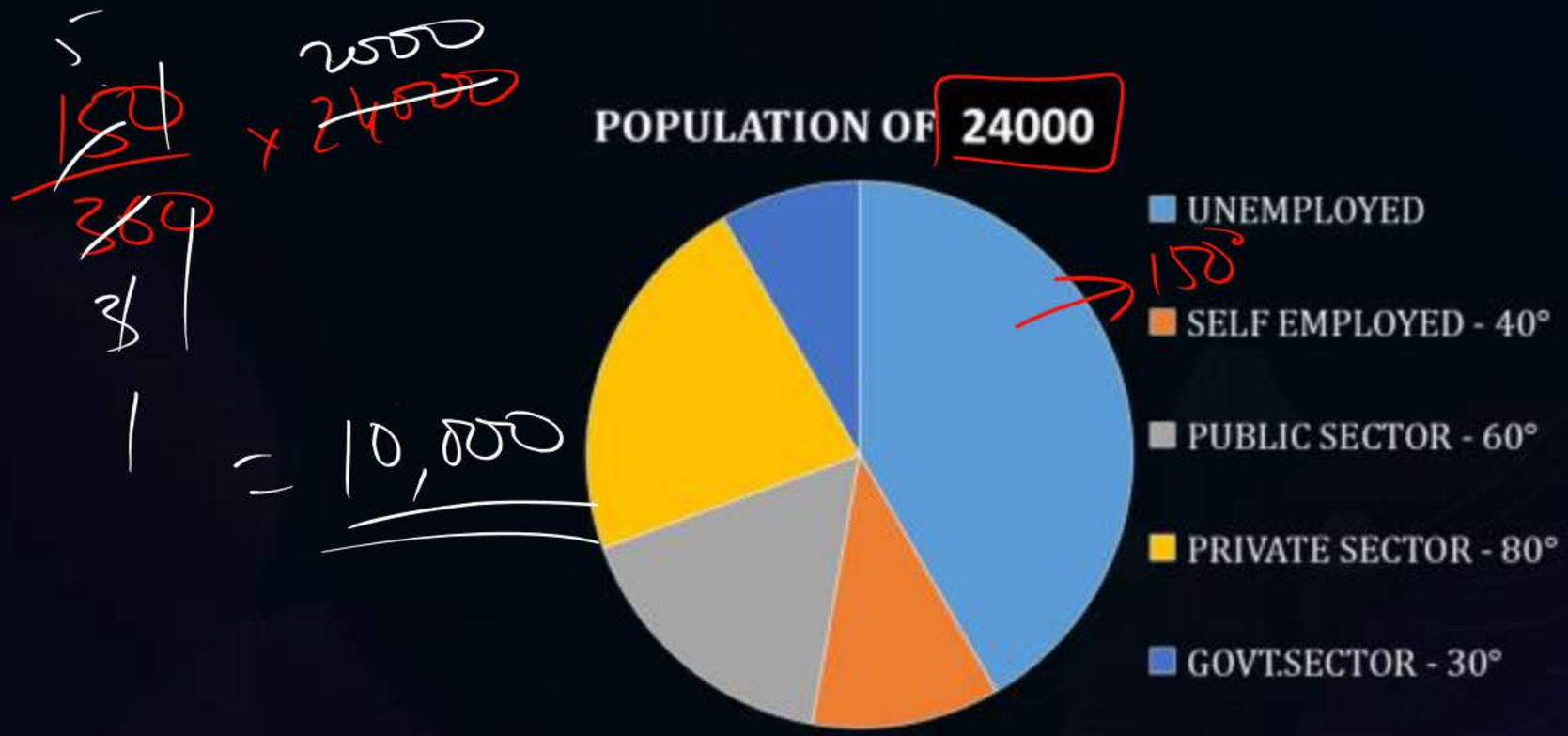
$$= \frac{100}{7}\%$$

$$= 14.28\%$$



[MCQ]

#Q. Find the number of people who are unemployed.



Tabular form:

Study the following table and answer the questions:

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]

#Q. Who among the following employees earns maximum bonus in comparison to his total income?

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A P

B Q

C R

D S

$P \Rightarrow \frac{80}{900} \times 100 = 8.\overline{8}\%$

$Q \Rightarrow \frac{40}{5}\% = 8\%$

$R \Rightarrow 10\%$

$S \Rightarrow \frac{80}{7} = 11.42\%$

$T \Rightarrow 10\%$

[MCQ]

#Q. The income from overtime is what percent of income from arrears in the case of employee P?

A. 90

B. 80

C. 40

D. 75

$\frac{180}{200} \times 100$

$= 90\%$

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]

#Q. How many employee have their salary more than 3 times the income from bonus?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A One

B Five

C Three

D None

[MCQ]

#Q. Who among the following employees has maximum percentage of his salary out of the total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A T

B P

C S

D R

[MCQ]

#Q. Who among the following employees has minimum ratio of income from arrears to the income from salary?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A P

B Q

C T

D S



2 mins Summary

Topic Data Interpretation

THANK - YOU